Databases 2.0

5th September 2024

Examination Paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

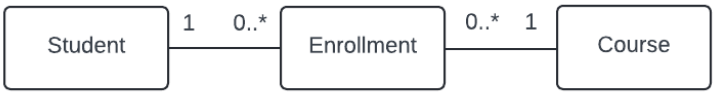
Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Question 1

A section of a university database system that manages student records and course enrolments is shown below.



a) Define what a primary key is. Provide ONE (1) example of a primary key **and** explain its significance in the context of the university database. **4**

b) Define what a foreign key is. Provide ONE (1) example of a foreign key **and** explain its role in the context of the university database. **4**

c) Explain TWO (2) advantages of normalisation. **2**

Total 10 Marks

Marks

Question 2

Identify **and** explain FIVE (5) functions a Relational DBMS performs. **10**

Total 10 marks

Marks

Question 3

10

Scenario:

A fitness centre offers various classes such as yoga, spinning and boxing. The system tracks which members are enrolled in which classes.

Draw an Entity Relationship Diagram to represent the fitness centre's class management system described above. You need to identify entities, relationships, cardinalities and label all the primary and foreign keys. Clearly state which entity the keys belong to.

Total 10 marks

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Marks

Question 4

a) A car rental company manages its fleet of cars, customer reservations and maintenance records. Each car can be rented by multiple customers over time and each customer can make multiple reservations. Maintenance records track the service history of each car. **4**

Entities:

- Car
- Customer
- Reservation
- Maintenance

Create a CRUD matrix to show the following transactions:

Transaction 1 – Add a new car to the fleet

Transaction 2 – Create a new reservation for an existing customer

Transaction 3 – Remove a car from the fleet and all its maintenance records

Transaction 4 – Update a car's maintenance record after a service

b) A number of business rules have been defined for the car rental company's database system as follows: **6**

Rule 1: All cars must have a unique identification number.
Rule 2: Reservations cannot be made for cars that are currently under maintenance.
Rule 3: Each maintenance record must include the date of service and the details of the service performed.

Discuss how each might be enforced on the system.

Total 10 marks

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Marks

Question 5

a) A local library maintains a spreadsheet to track book borrowings, recording each borrower's ID and name along with the titles and due dates for books borrowed. Each borrower can borrow multiple books, but the library holds only ONE (1) copy of each book. The current structure of the data does not conform to the First Normal Form (1NF). Convert the provided data into 1NF. Make sure you clearly identify keys. **8**

Provided data:

borrower_id	borrower_name	borrowed_books
101	Bob Smith	"The Pragmatic Programmer" due 2024-10-01 "Intro to DB Systems" due 2024-10-08
102	Alice Brown	"Python Crash Course" due 2024-12-15
103	Jane Doe	"The Art of Computer Programming" due 2024-09-23 "Data Science for Dummies" due 2024-10-31

b) A small business has been using flat files to manage its data but is considering switching to a database system. Identify TWO (2) advantages of using a relational database system over flat file. **2**

Total 10 marks

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Marks

Question 6

Consider the data in the following `movie` table. Write the appropriate SQL command to perform each of the following tasks.

movie

movie_id	movie_title	rental_price	max_rental_days
1	Inception	8.99	5
2	Toy Story	5.99	7
3	Avatar	7.99	6

a) Create the `movie` table. **6**

b) Update the `movie` table to include a new column named `rental_date` to record the date of the last rental. **2**

c) Remove the `max_rental_days` column in the movie table. **2**

Total 10 marks

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Marks

Question 7

Scenario: Technology Retailer Database

A technology retailer has set up a database to manage information about its products, customers, and orders. Below are the tables with sample data.

Product

product_id	product_name	category_id
1	laptop	1
2	smartphone	2
3	headphones	3
4	monitor	1

Category

category_id	category_name
1	Computers
2	Mobile Devices
3	Accessories

Order

order_id	product_id	order_date
1	1	2024-06-23
2	2	2024-07-02
3	3	2024-09-14
4	2	2024-09-15
5	4	2024-09-16

a) Write an SQL command to retrieve a list of all products along with their category names. Include both the product's name and the category name in your results. **3**

b) Write an SQL command to find the total number of orders placed for each product. The result should list the product name and the total number of orders. **3**

c) Write an SQL command to list the names of the products that have been ordered more than once. Include the product names and the number of times they have been ordered in your results. **4**

Total 10 marks

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Marks

Question 8

A small restaurant has been using manual data recording system to manage its inventory and customer orders. The restaurant tracks information such as ingredient stock levels, supplier details and daily orders in physical logbooks. The restaurant is considering switching to a relational database system to improve data management. **10**

Identify FIVE (5) advantages of using a relational database system over a manual system. Provide ONE (1) specific example to illustrate each advantage.

Total 10 marks

Marks

Question 9

a) A medium-sized online retail company has experienced several security incidents recently involving their database. **6**

Identify THREE (3) areas of loss that the company might face due to inadequate database security. Provide a brief explanation of each area of loss.

b) Identify TWO (2) models of Cloud databases and describe each model. **4**

Total 10 marks

Marks

Question 10

Data models are abstract representations of the data structures and the relationships between data elements. Identify **and** explain FIVE (5) different data models. **10**

Total 10 marks

End of Paper

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